

# Michael DeHart

Unreal Engine C++ / Gameplay Systems Programmer

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Self-directed C++ systems programmer specializing in UE networking, gameplay frameworks, and UI. Built custom equivalents of Data-Registry and CommonUI, a net bandwidth-optimized replication pipeline, and a data-driven gameplay framework that is similar to Epic's Modular Gameplay. Deep, hands-on knowledge of both legacy and Iris replication.

## PROJECTS

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**Invokers** Multiplayer FPS RPG · UE 5.4 (Modified Source Build) · Solo, C++ Dec 2023 – Present

- **Modular gameplay framework** — Bare-bones core actors (GameMode, GameState, PlayerController, PlayerState, Character, Pawn) that derive all behavior from net-replicated components selected by a GameplayTag which allow entire game modes to be completely data-driven. Architecture works similarly to Epic's Modular Gameplay but was built from scratch in parallel with our data manager.
- **4-byte network handle** — Engineered a compact DataTable row-handle that net-replicates any item or data reference at a constant 4 bytes, which is ~90% smaller than soft-object-path replication. Heavy data resolves locally using the 4 byte unique Id within our custom built data registry.
- **Custom UI framework** — a CommonUI-style framework written directly on hand-written Slate (not the UMG designer). Activatable widget stacks, layer & layout management, and custom input routing.
- **Replicated inventory** — Multiple inventory types (equipment, crafting, storage) over Fast Array Serialization, syncing only per-item deltas as 4-byte handles + 2 byte position data. — Inventory updates cost ~6 bytes × changed items regardless of item complexity.
- **Networked GameMode transitions** — a custom replication layer with a seamless-travel handshake that is gated until all clients acknowledge changes or they are kicked by a configurable timeout.

**Paranoia** Single-player Psychological Horror · UE 5.2 · Solo, C++ Apr 2023 – Dec 2023

- **Centralized event dispatcher** — A custom C++ system driving player tasks, entity spawning, and scripted scenarios from data-defined attributes, keeping game flow decoupled and designer-editable.

## TECHNICAL SKILLS

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**LANGUAGES** C++ (primary) · C# · Lua · Blueprint · JavaScript / TypeScript · React

**UNREAL** GAS · Modular Gameplay · Slate · UMG · CommonUI · Gameplay Tags · DataTables · Mass Entity · AI Perception · Chaos · UnrealEd · PropertyEditor

**NETWORKING** Legacy · Iris · FAS · Custom NetSerialize · Subobject Replication

**TOOLS** Git · Perforce · Diversion · Visual Studio · Unreal Insights · Network Profiler

## EXPERIENCE

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**IT Project Manager** - Wintech Technology Inc Aug 2013 – Jan 2026

- Led IT teams and built uptime-critical infrastructure for 12+ years, managed client relationships through long-term project delivery and held workshops to train other IT professionals.

**SEO Specialist** - Linkflow Apr 2021 – Aug 2022

- Built and executed SEO strategy for major brands, such as Logitech and MyFitnessPal, by growing their organic traffic through tailored link acquisition and SEO optimized site modifications.